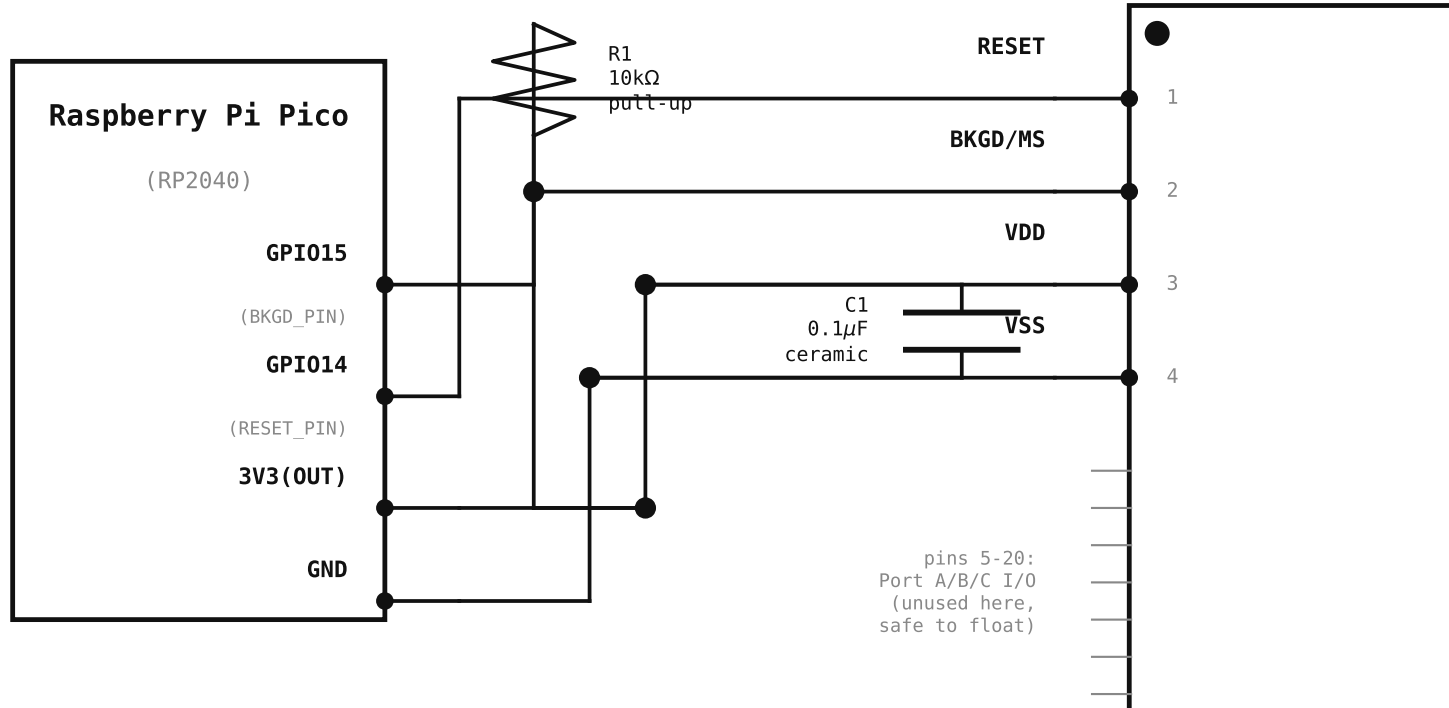


pico-bdm -- Raspberry Pi Pico to MC9S08SG8 / MC9S08SG4 Wiring

20-Pin TSSOP pinout confirmed against MC9S08SG8 datasheet Table 2-1 / Figure 2-1

MC9S08SG8 / MC9S08SG4

20-Pin TSSOP -- S9S08SG8E2MTJ



Notes

1. Power the target at 3.3V only -- do NOT connect to 5V. Pico GPIOs are 3.3V and not 5V-tolerant.
2. R1 (10k pull-up, BKGD to VDD) is extra robustness on top of the chip's own internal BKGD pull-up.
3. C1 (0.1μF ceramic) must sit as close to VDD/VSS (pins 3/4) as physically possible -- datasheet requirement, not optional.
4. A larger bulk capacitor (~10μF) on the 3.3V rail is recommended if the wire run from the Pico is long.
5. RESET alone cannot force BDM mode -- BDM is entered by holding BKGD low across a RESET pulse (see firmware/bdc.py: hardware_reset_to_bdm()). This is confirmed in datasheet section 2.2.3.
6. Pins 5-20 (Port A/B/C I/O) are unused for this bring-up test and safe to leave floating.